Vector-Shield: UI Testing & Verification Guide

Target App: https://vectorshield.streamlit.app/ | **Author:** VigneshVS2005 | **Hardware:** NVIDIA A100 GPU

1. Executive Summary & Proof of Authenticity

Vector-Shield is an enterprise neuro-symbolic security middleware defending RAG pipelines from Indirect Prompt Injections. It is 100% authentically running on an NVIDIA A100 GPU: sentence-transformers (384-D), Scikit-Learn LOF, and 149M-parameter ModernBERT weights (598 MB in models/stage2_modernbert/).

2. General Test Case: Tab 1 — Live Attack Simulator

Test A (Shield OFF): Select "Direct Financial Hijack". Turn Enable Vector-Shield Defense OFF. Click Execute Agent Query. Output shows EXPLOIT EXECUTED: the naive agent authorizations a fraudulent ■1,50,000 transfer.

Test B (Shield ON): Keep the same query. Turn Vector-Shield Defense ON. Click Execute Agent Query. System calculates LOF score (>2.00) in 1.8ms, isolates the outlier, triggers LangGraph self-healing, and outputs a secure sanitized response.

3. Dynamic Document Ingestion Test: Tab 3 (Real PDFs)

Real sample PDFs are provided directly in **data/sample_documents/**:

• **poisoned_vendor_agreement.pdf:** Contains hidden [SYSTEM DIRECTIVE] to bypass transfer limits. Ingestion fails with ■ BLOCKED & QUARANTINED (LOF > 1.85).

• **clean_banking_sop.pdf:** Contains standard KYC and compliance protocols. Ingestion succeeds with ■ APPROVED & INDEXED (LOF < 1.20) and increments active doc count.

4. What is SIEM & Live Webhook Verification: Tab 6

SIEM (Security Information & Event Management): Enterprise platforms (Splunk, Microsoft Sentinel, Datadog) used by Security Operations Centers (SOC) to monitor real-time cyber threats. Vector-Shield broadcasts structured JSON alert webhooks upon every quarantined injection.

How to test live right now:
1. Open **https://webhook.site** in a browser to generate a free real-time webhook URL.
2. Paste that URL into Tab 6 in the Vector-Shield dashboard.
3. Click "Dispatch Test SIEM Alert".
4. Switch back to Webhook.site and observe the real HTTP POST request with full incident JSON payload arrive live.

5. Quick Reference for Remaining Tabs

• **Tab 2 (ModernBERT):** Real forward pass on A100 GPU classifying text into Benign vs Injection probabilities.
• **Tab 4 (3D Hyperspace):** Interactive 3D PCA plot showing green corporate clusters vs red quarantined outliers.
• **Tab 5 (Topology):** Mathematical formulation and sub-2ms latency comparison.

Generated autonomously on NVIDIA A100 GPU Node | Vector-Shield Project Repository:
https://github.com/VigneshVS2005/Vectorshield